Problem 5.4

(a) Show that Banker's Rule is usually more favorable to the lender than ordinary simple interest.

(b) Give an example in which the opposite relationship holds.

Let P be the principal and r the annual interest rate.

(a)

Under Banker's Rule, the interest is computed with the exact number of days over a 360-day year:

$$I_{\text{banker}} = Pr\left(\frac{d_{\text{actual}}}{360}\right).$$

Under ordinary simple interest, the interest is computed with approximate time, using 30 days for each month, also over a 360-day year:

$$I_{\text{ordinary}} = Pr\left(\frac{d_{\text{approx}}}{360}\right).$$

Usually, the actual number of days is greater than the approximate number of days because actual time counts months with 31 days, while ordinary simple interest treats each month as having only 30 days. Thus,

$$\frac{d_{\text{actual}}}{360} > \frac{d_{\text{approx}}}{360}.$$

Multiplying by the positive quantity Pr, we get

$$I_{\text{banker}} > I_{\text{ordinary}}.$$

Therefore, Banker's Rule is usually more favorable to the lender than ordinary simple interest.

(b)

The opposite can happen when the time period includes February.

For example, suppose the investment lasts exactly one month in February of a non-leap year. Then ordinary simple interest uses 30 days, while Banker's Rule uses the actual 28 days:

$$\frac{30}{360} > \frac{28}{360}.$$

Hence,

$$Pr\left(\frac{30}{360}\right) > Pr\left(\frac{28}{360}\right),$$

so in this case ordinary simple interest is greater than Banker's Rule.